Share -ware !

# Simplified Aircraft Design Spreadsheet f

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LEGAL NOTICE !

Better to save this spreadsheet as a renamed copy before changing it for your a

## Wing Loading

| Inputs | | | Calculated Values | | |
|---|---|---|---|---|---|
| Stall speed | (kts) | 60 | Stall speed | (ft/sec) | 101.3 |
| Takeoff air density | (slugs/ft^3) | 0.00238 | Dynamic pressure | (psf) | 12.2 |
| Wing CLmax | | 1.6 | Wing loading (W/S) | (psf) | 19.55 |
| power loading | (lb/hp) | 8.3333 | | | |

## Wo -known engine

| | | | | | |
|---|---|---|---|---|---|
| *If sizing to a range requirement, ignore this input for now but enter it later when Wo is calculated and engine is selected* | | | *You cannot enter Wo! Engine power and power loading are multiplied to calculate Wo. If sizing to a range requirement, do calculation below, read Wo from graph, and select an engine that gives the desired Wo=P*P/W* | | |
| Engine Power | (hp -each) | 120 | Wo | (lb) | 2000.0 |
| Number of Engines | | 2 | Wing Area | (sq ft) | 102.3 |

## Wo Sizing to a range requirement (engine is not yet selected)

| | | | | | |
|---|---|---|---|---|---|
| Swet/Sref | | 4.2 | | | |
| Cfe | | 0.0053 | Cdo | | 0.0223 |
| Aspect ratio (A) | | 10 | K (=1/piAe) | | 0.0424 |
| Cruise air density | (slugs/ft^3) | 0.00176 | W/S cruise | | 19.2 |
| Cruise velocity | (kts) | 180 | Cruise velocity | (ft/sec) | 304.0 |
| | | | Dynamic pressure | (psf) | 81.3 |
| | | | L/D cruise | | 9.57 |
| Engine SFC | (lb/hour /bhp) | 0.45 | Engine SFC | (lb/sec /bhp) | 0.000125 |
| Prop Efficiency (cruise) | | 0.75 | | | |
| Range | (nmi) | 800 | Range | (ft) | 4860800 |
| | | | Breguet Exponent | | 0.1539 |
| | | | Wf/Wo | | 0.1641 |
| Fuel allowance | (%) | 6 | Wf/Wo with allow. | | 0.1739 |
| Empty Weight constant | "a" | 1.2438 | | | |
| Weight - crew | (lbs) | 180 | | | |
| Weight - Passengers | (lbs) | 180 | | | |
| Weight - payload | (lbs) | 20 | See Sizing Graph sheet for Wo Results | | |

## Wing Geometry

| | | | | | |
|---|---|---|---|---|---|
| Wing taper ratio | | 0.5 | Wing Span | (ft) | 31.98 |
| | | | Root Chord | (ft) | 4.26 |
| | | | Tip Chord | (ft) | 2.13 |
| | | | Mean Chord | (ft) | 3.32 |

## Tail Sizing

| | | | | | |
|---|---|---|---|---|---|
| Horizontal tail arm | (ft) | 10 | Tail areas: | | |
| Cht (volume coeff) | | 0.6 | Sht (horizontal) | (sq ft) | 20.35 |
| Vertical tail arm | (ft) | 10 | | | |
| Cvt (volume coeff) | | 0.04 | Svt (vertical) | (sq ft) | 13.08 |

# or Homebuilders

*e legal notice for further restrictions.*

airplane, in case it gets messed up!

**Enter items in blue and purple. Come back and re-enter purple items after the drawing is analyzed.**

Send problems & questions to SADFH@aircraftdesig

| Other Factors Used | Equations (from book) | Misc Useful Calcs |
|---|---|---|

(OK to change these if you know what you are doing)

$$q = \frac{1}{2}\rho V_{Stall}^2$$

$$\frac{W}{S} = qC_L$$

$$C_{D0} = C_{fe}\frac{S_{wet}}{S_{ref}}$$

| | |
|---|---|
| e (Oswald) | 0.75 |
| Wcruise/Wo | 0.98 |

$$K = \frac{1}{0.75\,\pi\,A} = \frac{0.424}{A}$$

| | |
|---|---|
| CL-cruise | 0.2356 |

$$\frac{L}{D} = \frac{1}{\frac{qC_{D0}}{W/S} + (W/S)\frac{K}{q}}$$

| | |
|---|---|
| Non-cruise weight allowance | 0.975 |

$$W_f/W_0 = 1 - 0.975\,e^{\frac{-R\,c_{bhp}}{500\eta_p L/D}}$$

| | |
|---|---|
| Empty Weight exponent | -0.09 |

$$W_E/W_0 = aW_{0^{-0.09}}$$

| | | |
|---|---|---|
| Fuel Weight | 347.8 | (lbs) |
| Fuel Vol | 58.0 | (gal) |

$$W_0 = \frac{W_{people} + W_{payload}}{1 - W_E/W_0 - W_F/W_0}$$

*Preliminary tail sizing using these methods does not guarantee that the aircraft will be stable, controllable, or safe. You must perform stability calculations after doing your design layout!*

gn.com

## Sizing Calculations

| Wo guess | We/Wo | We | Wo calculated |
|---|---|---|---|
| 1000 | 0.6680 | 668.0 | 2403.2 |
| 1500 | 0.6440 | 966.0 | 2087.3 |
| 2000 | 0.6276 | 1255.1 | 1914.2 |
| 2500 | 0.6151 | 1537.7 | 1801.0 |

Pick Wo from graph, where the two lines cross. Enter this value below to find the minimum horsepower engine for your power loading.

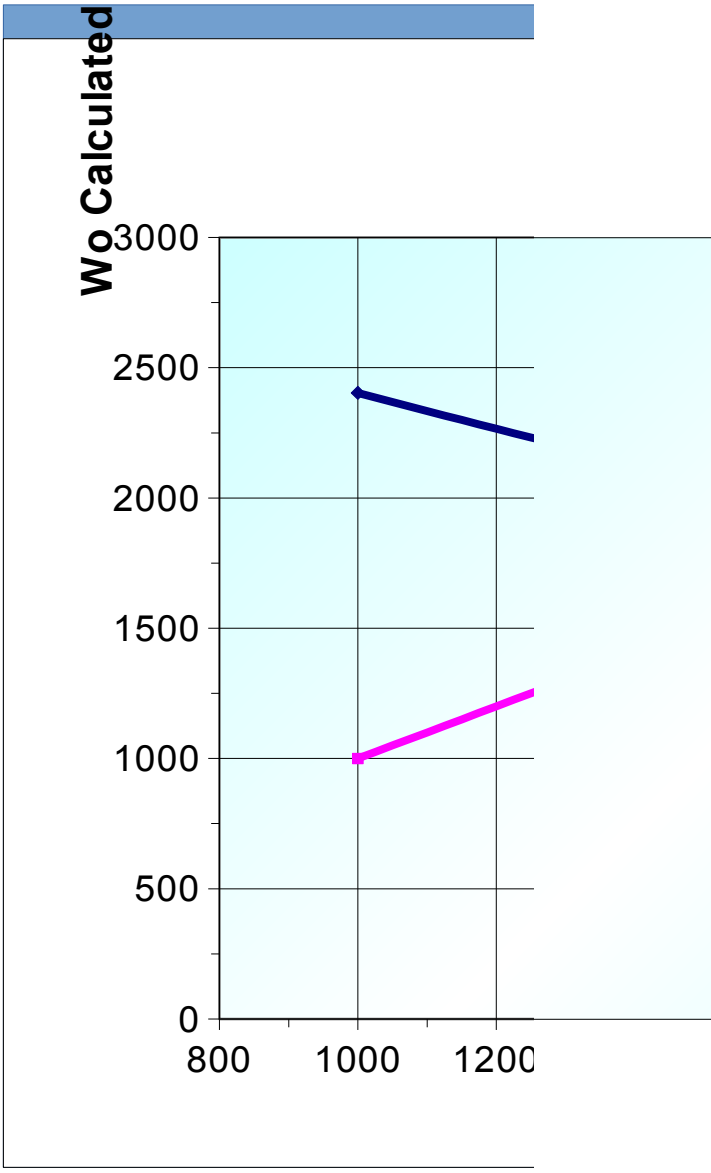
If sizing graph lines do not cross, change Wo-guess values above.

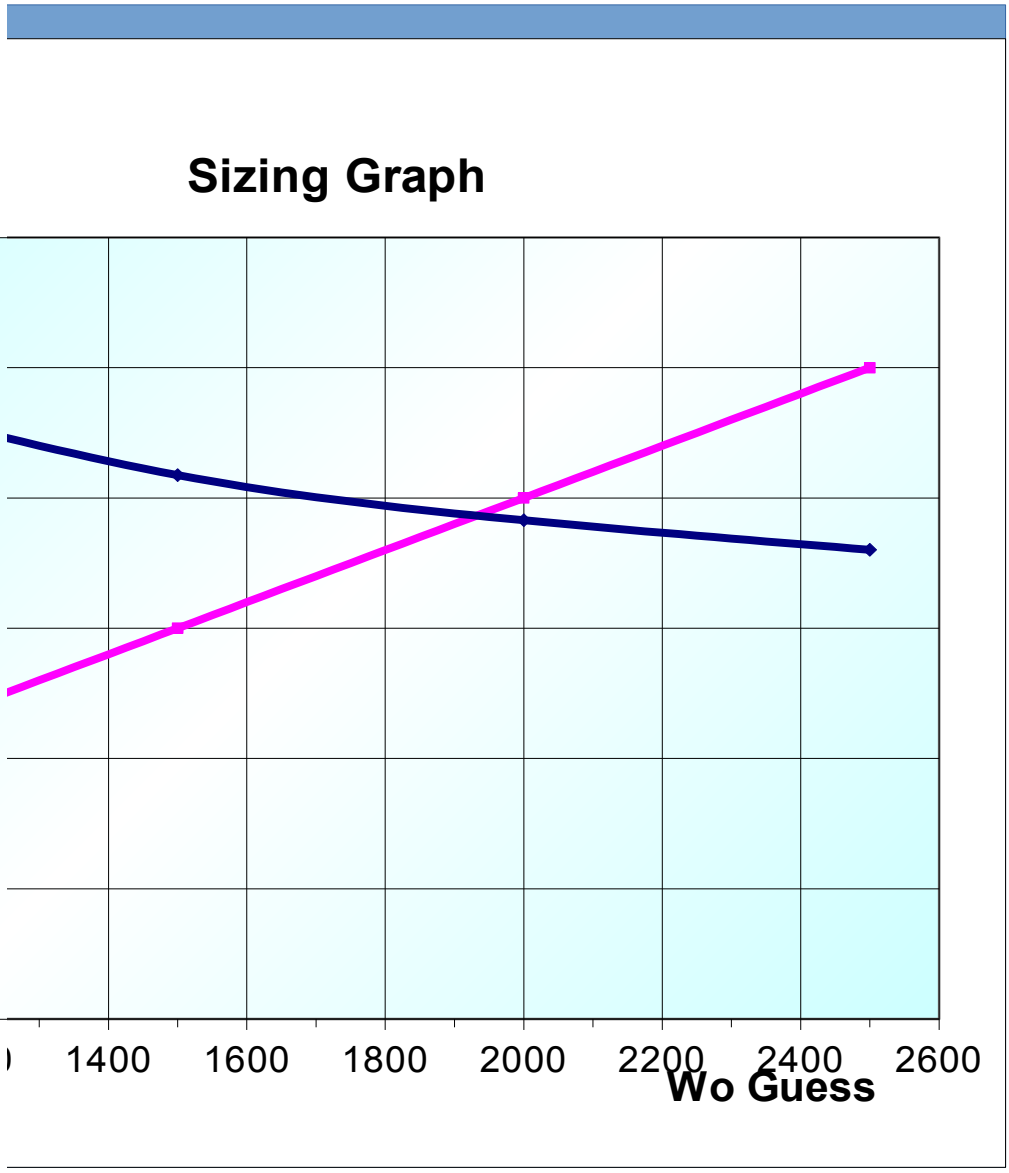
| | |
|---|---|
| Enter Wo from graph (lbs) | 2000 |
| Pick engine with horsepower of at least: | 120 |

Now find a suitable engine of at least this horsepower and enter its power below:

| | |
|---|---|
| Power of Selected Engine: | 120 |
| Calculated Power Loading: | 8.33 |

Now go to sheet 1 and enter the power of your selected engine and the power loading calculated above in the boxes this color.




Sizing Graph
1400
1600
1800
2000
2200
2400
2600
Wo Guess

# As-Drawn Performance Calculations for Simplified Aircraft Design fo

Use this sheet for performance calculations after you have drawn your design and measure

| Values from Sheet 1 - OK to change these here, but then link from Sheet 1 is lost | | |
|---|---|---|
| Wo | (lb) | 2000 |
| Engine Power | (hp) | 120 |
| Number of Engines | | 2 |
| Engine SFC | (lb/hour /bhp) | 0.45 |
| Wing Area | (sq ft) | 102.3 |
| Cdo | | 0.0223 |
| K (=1/piAe) | | 0.0424 |
| Wing CLmax | | 1.6 |
| Takeoff air density | (slugs/ft^3) | 0.00238 |
| e (Oswald) | | 0.75 |
| Wcruise/Wo | | 0.98 |
| Non-cruise weight allowance | | 0.975 |
| Fuel allowance | (%) | 6 |
| Weight - crew | (lbs) | 180 |
| Weight - passengers | (lbs) | 180 |
| Weight - payload | (lbs) | 20 |
| Empty Weight | (lbs) | 1255.1 |

| Other Inputs | | |
|---|---|---|
| Propeller Diameter | (ft) | 5 |
| Engine RPM | (rev/min) | 2700 |
| Cooling Power Loss | (%) | 6 |
| Cruise Power Setting | (% of SL hp) | 62 |
| Cruise air density | (slug/ft^3) | 0.00176 |

| Calculated Values | | |
|---|---|---|
| wing loading (W/S) | (psf) | 19.55 |
| stall speed | (ft/sec) | 101.34 |
| stall speed | (kts) | 60.00 |
| power loading | (lb/hp) | 8.33 |
| Takeoff Parameter | | 123.2 |
| Takeoff Groundroll | (ft) | 874.2 |
| Takeoff to 50 ft | (ft) | 1159.1 |
| Cruise Weight | (lb) | 1959.9922 |

| | | |
|---|---|---|
| Power Coefficient Cp | | 0.1317 |

Go to next sheet for climb, cruise, and maximum speed calculations then return here

| | | |
|---|---|---|
| Cruise speed | (kts) | 180 |
| Cruise Prop efficiency | | 0.85 |

| | | |
|---|---|---|
| Cruise speed | (ft/sec) | 304.02 |
| Cruise Advance Ratio J | | 1.3512 |
| Cruise q | (psf) | 81.3 |
| Cruise W/S | (psf) | 19.2 |
| Cruise L/D | | 9.6 |
| Wfuel (total) | (lbs) | 365 |
| Wfuel (usable) | (lbs) | 344 |
| Wfuel (cruise) | (lbs) | 294 |
| log term | | 1.1776909 |
| Range | (ft) | 5854552 |
| **Range** | **(nmi)** | **964** |

**r Homebuilders**

d its geometry.

**Equations (from book)**

$$T.O.P.=1.21\frac{\frac{W}{S}\frac{W}{hp}}{C_{L\max}}$$

$$\frac{L}{D}=\frac{1}{\frac{qC_{D0}}{W/S}+(W/S)\frac{K}{q}}$$

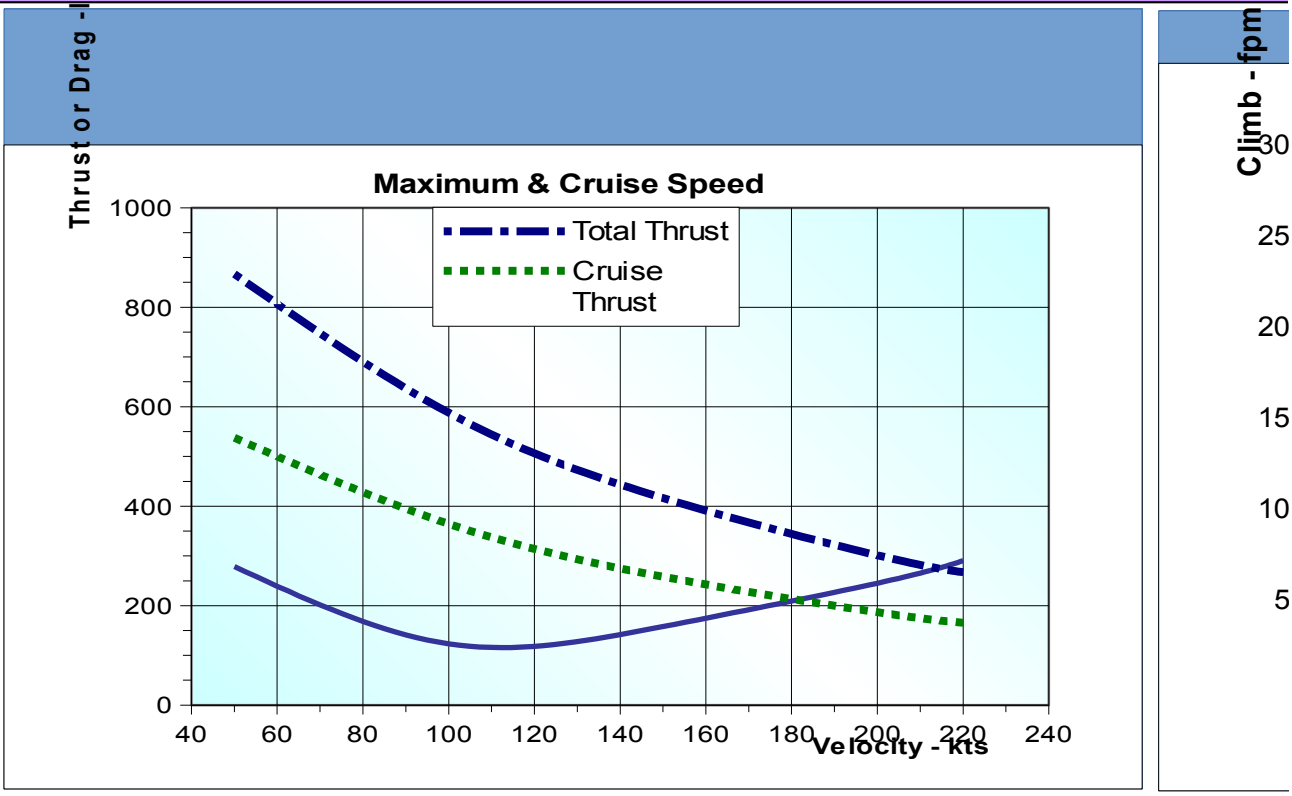
$$R=\frac{550\,\eta_p}{C_{bhp}}\frac{L}{D}\ln\left[\frac{0.975\,W_0}{W_0-W_{f(usable)}}\right]$$

## Climb, Cruise, & Max Speed Calculations for Simplified Aircraft Desi

Use this sheet for performance calculations after you have drawn your design and mea

| | | | Input adjusted values | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| V kts | V ft/sec | Advance Ratio J | Propeller Efficiency | Total Thrust lbs | Cruise Thrust lbs | Dynamic pressure q | CL | CD | Drag lbs |
| 50 | 84.45 | 0.375333 | 0.59 | 866.8703 | 537.4596 | 6.2759862 | 3.115636 | 0.434233 | 278.7433 |
| 100 | 168.9 | 0.750667 | 0.8 | 587.7087 | 364.3794 | 25.1039448 | 0.778909 | 0.048008 | 123.2702 |
| 150 | 253.35 | 1.126 | 0.85 | 416.2937 | 258.1021 | 56.4838758 | 0.346182 | 0.027346 | 157.9862 |
| 200 | 337.8 | 1.501333 | 0.82 | 301.2007 | 186.7444 | 100.415779 | 0.194727 | 0.023869 | 245.1549 |
| 220 | 371.58 | 1.651467 | 0.8 | 267.1403 | 165.627 | 121.503093 | 0.160932 | 0.023359 | 290.2979 |

sheet



**ign for Homebuilders**

sured its geometry.

**Equations (from book)**

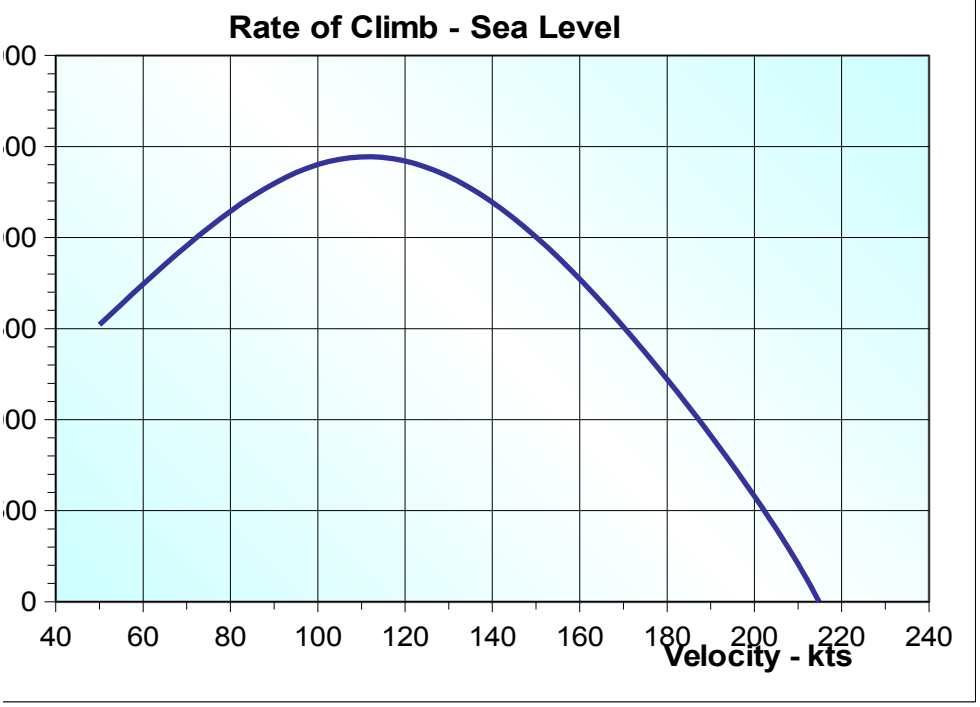
$$D = qS\left(C_{D0} + KC_{L^2}\right)$$

| Climb (fps) |
|---|
| 1520.4345 |
| 2401.3464 |
| 2003.3404 |
| 579.56153 |
| -263.4165 |

$$C_{L-cruise} = \frac{W/S}{q}$$

$$q = \frac{1}{2}\rho V^2$$

$$Vv = V\left[\frac{T}{W} - \frac{1}{L/D}\right]$$



**Weight Reporting Format for Simplified Aircraft Design for Homebuilders**

Use this sheet for reporting weights calculations & estimating center of gravity.

***Input only items i***

(Sorry, no weights estimation methods here, just the form to enter your estimates. Moments & cg are calculated)

| | Weight lbs | Loc ft | Moment ft-lbs |
|---|---|---|---|
| **STRUCTURES** | **661.0** | | **5600** |
| Wing | 276 | 6.5 | 1794 |
| Horizontal Tail | 24 | 21.0 | 504 |
| Vertical Tail | 18 | 19.0 | 342 |
| Ventral Tail | 8 | 17.0 | 136 |
| Fuselage | 155 | 9.0 | 1395 |
| Canopy | 15 | 8.0 | 120 |
| Nacelle on wing | 50 | 9.0 | 450 |
| Nacelle/cowling | 30 | 7.5 | 225 |
| Motor Mount | 10 | 7.5 | 75 |
| Main Landing Gear | 56 | 9.0 | 504 |
| Nose Landing Gear | 19 | 2.9 | 55 |
| **PROPULSION** | **411.0** | | **2877** |
| Engine | 340 | 7.0 | 2380 |
| Air Induction | 3 | 7.0 | 21 |
| Cooling | 3 | 7.0 | 21 |
| Exhaust | 8 | 7.0 | 56 |
| Engine Controls | 2 | 7.0 | 14 |
| Misc. Engine Inst | 5 | 7.0 | 35 |
| Propeller | 30 | 7.0 | 210 |
| Starter | 10 | 7.0 | 70 |
| Fuel System | 10 | 7.0 | 70 |

| | Weight lbs | Loc ft | Moment ft-lbs |
|---|---|---|---|
| **EQUIPMENT** | **69.0** | | 429 |
| Flight Controls | 10 | 5.5 | 55 |
| Instruments | 10 | 5.5 | 55 |
| Hydraulics | 2 | 6.0 | 12 |
| Electrical | 12 | 6.0 | 72 |
| Avionics | 15 | 5.0 | 75 |
| Air Conditioning | | | 0 |
| Anti-Icing | | | 0 |
| Furnishings & Equipment | 20 | 8.0 | 160 |
| (% We Allowance) | 10 | | |
| Empty Weight Allowance | 114.1 | 7.8 | 891 |
| **TOTAL WEIGHT EMPTY** | **1255.1** | **7.8** | 9797 |
| **USEFUL LOAD** | **744.9** | | |
| Crew | 180.0 | 8.0 | 1440 |
| Fuel | 358.9 | 7.5 | 2692 |
| Oil | 6 | 5.0 | 30 |
| Passengers | 180 | 8.0 | 1440 |
| Payload | 20 | 10.0 | 200 |
| **TAKEOFF GROSS WEIGHT** | **2000.0** | **7.8** | 15598 |

enter %

don't enter these

Calculated Empty Wei
to Initial Sizing Inputs s
in purple box:

Don't enter this! It is calcu

Possible Loading Conditions

| | | | |
|---|---|---|---|
| Crew+Pass+Pld, No Fuel | 1641.1 | 7.9 | 12907 |
| Crew+Pass,No Pld,No Fuel | 1621.1 | 7.8 | 12707 |
| Crew only, No Fuel | 1441.1 | 7.8 | 11267 |
| Crew only, Full Fuel | 1800.0 | 7.8 | 13958 |

***n blue***

| | |
|---|---|
| ght constant "a". Go sheet and revise value | 1.2438 |

ulated from Wo, We, and other Useful Load Group items

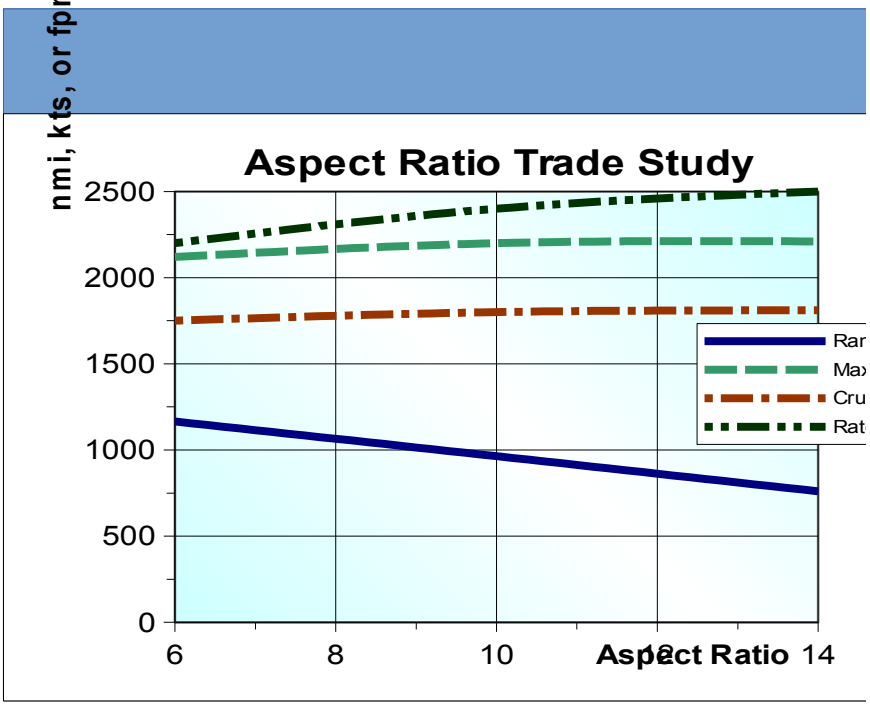
# ASPECT RATIO OPTIMIZATION: DR-4

| | |
|---|---|
| A-baseline | 10 |
| Wo | **2000.0** |
| Wwing-base | 276 |
| We-base | **1255.1** |

Read performance results and enter here

| A | A/Abase | Wwing/ Ww-base | Delta We | We-new | Range | Max Speed | Cruise Speed | ROC |
|---|---|---|---|---|---|---|---|---|
| 6 | 0.60 | 0.74 | -73 | 1182 | 1165 | 212 | 175 | 2200 |
| 10 | 1.00 | 1.00 | 0 | 1255 | 964 | 220 | 180 | 2400 |
| 14 | 1.40 | 1.22 | 62 | 1317 | 760 | 221 | 181 | 2500 |

Enter delta on Weight sheet (I stuck it in the unused Anti-Ice entry), and change aspect ratio and empty weight "a" factor on Initial Sizing Sheet



| max*10 | cruise*10 |
|---|---|
| 2120 | 1750 |
| 2200 | 1800 |
| 2210 | 1810 |

